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Remaining useful life prediction method based on stacked autoencoder and generalized wiener process for degrading bearing

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Abstract

Remaining useful life (RUL) prediction using deep learning networks primarily produces point estimates of RUL, but capturing the inherent uncertainty in RUL prediction is difficult. The use of the stochastic process approach can reflect the uncertainty in RUL predictions. However, the amount of data generated during equipment operation cannot be effectively utilized. This paper aims to propose an adaptive RUL prediction method tailored for extensive datasets and prediction uncertainty, effectively harnessing the strengths of deep learning methods in managing massive data and stochastic process techniques in quantifying uncertainties. RUL prediction method, based on stacked autoencoder (SAE) combined with Generalized Wiener Process, employs SAE to extract profound underlying features from the monitoring signals. Principal component analysis (PCA) is then used to select highly trending features as inputs. The output of PCA accurately reflects health status. A Generalized Wiener Process is used to construct a model for the evolution of the health indicators. The estimation values for the model parameters are determined using the Maximum Likelihood Estimation method. Furthermore, an adaptive update is performed based on Bayesian theory. Utilizing the sense of the first hitting time concept, the Probability Density Function for RUL prediction is derived accurately. Finally, the effectiveness and superiority of the proposed method is verified using numerical simulations and experimental studies of bearing degradation data. The method improves the life prediction accuracy while reducing the prediction uncertainty.

Keywords: remaining useful life prediction, stacked autoencoder, generalized wiener process, degradation modeling, feature extraction

1. Introduction

Rolling bearings are crucial components of rotating mechanical equipment, playing a vital role in supporting the

high-precision rotation of mechanical rotating parts [1]. As the operating time prolongs, bearings may undergo deterioration in their performance and health status caused by diverse internal and external factors. Once a failure occurs, it significantly impacts the safety and reliability of mechanical equipment. Therefore, accurately predicting the remaining useful life (RUL) of bearings and scheduling maintenance plans

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based on RUL are of significant practical importance. This approach enhances the reliability of bearing operation and ensures the safe functioning of equipment.

Currently, RUL prediction methods can be broadly categorized into three main types: based on failure mechanisms, data-driven methods, and hybrid approaches that integrate both [2]. Methods grounded in failure mechanisms depend on precise mathematical descriptions of system degradation, facilitating precise equipment RUL predictions. However, as equipment becomes increasingly complex, nonlinear, and high-dimensional, accurately modeling the evolution patterns of its health status becomes challenging. Obtaining economical failure mechanism models proves daunting, thus curtailing the widespread adoption of these methods to some extent. Data-driven methods, relying on sensor-acquired data, utilize machine learning and probabilistic statistical techniques to uncover implicit equipment health information within the data. The development of data-driven RUL prediction methods has emerged as the current mainstream and research focus. Data-driven approaches primarily involve three steps: data collection, establishing health indicators (HIs), and lifespan prediction. HI are derived from collected data, extracting meaningful features to quantitatively describe the equipment's degradation process. Appropriate HIs aid in simplifying prediction models and achieving more accurate forecasts. Hence, determining clear HIs stands as a pivotal challenge [3].

In order to describe the performance degradation process of devices more accurately and quantitatively, many scholars have devoted themselves to the research of HI construction methods. Gebrael *et al* [4] derived the HI of the bearing by extracting mean amplitude and harmonics from vibration data to forecast the RUL. Li *et al* [5] extracted the root-mean-square values from the vibration signals of the bearings and constructed the state transition matrices so as to predict the RUL of the bearings. In recent times, deep learning has become one of the most dynamic fields of research. By emulating the learning processes of the human brain and harnessing big data to discern intricate features, deep learning captures the inherent complexity of data, enhancing prediction accuracy significantly. Guo *et al* introduced a HI based on recurrent neural network for bearing RUL prediction. The two bearing datasets verify that HI has high monotonicity and correlation, which is favorable for bearing RUL prediction [6]. By improving the restricted Boltzmann machine, Liao *et al* employed a novel regularization factor to autonomously generate features and establish HIs using self-organizing mapping (SOM). On this basis, to minimize the reconstruction error of the deep belief network (DBN), Shao *et al* [7, 8] implemented a study of RUL prediction based on HI using continuous DBN. She *et al* [9] proposes a trend-based HI construction method using a sparse auto-encoder with regularization model for rolling bearings, improving RUL prediction. Experimental validation with PRONOSTIA and ABLT-1A data demonstrates the method's effectiveness, showcasing its potential for application in mechanical predictive maintenance. Yao *et al* [10] introduces a robust data-driven approach for accurate RUL prediction of rolling bearings, employing convolutional denoising autoencoder and bidirectional

long short-term memory network (LSTM) to mitigate the impact of noise in industrial data. Experimental validation on the Xi'an Jiaotong University bearing dataset confirms the method's effectiveness in achieving reliable RUL predictions. Han *et al* [11] achieved the necessary one-dimensional HI construction by enhancing the feature extraction capabilities of the SAE through increased neural network depth. Francisco *et al* introduced an integrated approach based on DAE and Local Linear Embedding to construct an HI that reflects the dynamic degradation of bearings, which was employed in their research for lifetime prediction. Experimental results on rolling bearings indicate that their method provides a more ideal prediction accuracy. Therefore, autoencoders are well-suited for modeling continuous monitoring data of deteriorating equipment [12]. Stacked autoencoders (SAEs) can be built using different types of neural networks. Due to their excellent performance in feature extraction, convolutional neural networks (CNNs) are widely used as the base neural network for SAEs. However, traditional CNN networks require a lot of computational resources due to their complex structure when handling high-resolution images and multi-channel data. The large number of convolutional kernels results in many parameters that need to be trained, increasing the model's storage requirements and training time. Additionally, when the data volume is small, the large number of parameters and high model complexity easily lead to overfitting issues. To address this issue, this paper uses separable CNNs as the base neural network, significantly reducing computational load through decomposing convolution operations and speeding up training and inference. The substantial reduction in the number of parameters is suitable for resource-constrained environments. It allows for more efficient use of computational resources, improving the overall efficiency and performance of the model. Additionally, the reduced number of parameters lowers model complexity, decreasing the risk of overfitting, enhancing the model's generalization ability and flexibility, and improving scalability. This approach significantly shortens computation time without compromising feature extraction performance. RUL prediction pertains to the future failure events of equipment. As equipment is subjected to various stochastic factors such as environmental conditions and loads during its service life, the predicted outcomes inevitably carry uncertainty. The statistics-driven approach is based on the theory of probabilistic statistics, modeling the evolution of performance degradation variables in the framework of stochastic models, and formulating the RUL distribution as probability distributions, which not only yields a point estimate of the RUL, but also describes the uncertainty of the prediction.

Wiener process is a category of diffusion process propelled by Brownian motion with independent increments and Gaussian distribution, making it apt for portraying non-monotonic degradation processes and providing more flexibility in modeling degradation measurement signals. Consequently, it finds extensive application in degradation modeling and RUL prediction for rolling bearings, lithium batteries, and lasers. Yan *et al* [13] modeled the degradation process and used it to predict the RUL of wheel treads of

high-speed trains using several related Wiener processes for degraded products with multiple performance characteristics. Duan *et al* [14] derived the expressions for the probability density function and the cumulative distribution function of the RUL distribution, taking into account the effects of random effects and measurement errors, using Wiener process modeling. Liu *et al* [15] developed a real-time RUL prediction model for a two-phase linear Wiener process. Li *et al* [16] proposed a Rolling Bearing RUL prediction method based on the Generalized Wiener process, addressing the nonlinearity of the neglected random error terms in traditional approaches. In an attempt to leverage the advantages of deep learning methods and statistical data-driven approaches, while addressing the limitations inherent in these approaches, researchers have conducted exploratory investigations. For instance, Deutsch *et al* [17] utilized DBNs to extract degradation features, followed by representation of the feature evolution trend through a stochastic degradation model. The Particle Filter algorithm was employed for model updates, yielding the numerical form of the RUL probability distribution. Hu *et al* [18] utilized DBN for extracting intricate degradation features from the high-dimensional vibration signals of bearings, input the HI into LLE based on the trend screening features, and used the diffusion process to model and derive the probability density function of RUL prediction. Furthermore, this method was enhanced in the form of CDBN combined with SOM and nonlinear diffusion processes [19]. Dai *et al* [20] proposed a HI construction model based on wavelet kernel networks combined with bi-directional gated cyclic units and a reliability assessment method based on the Wiener process. Although many scholars have done a lot of useful work, considering the variable operating environment of bearings, which leads to an extremely complex performance degradation process, when modeling the degradation of bearings at the degradation stage, the above literature methods all use white noise or linear time-varying functions to characterize their random error terms, which sometimes struggle to precisely depict the degradation process of bearing performance, thereby leading to notable prediction inaccuracies. This paper proposes an adaptive degraded bearing RUL prediction method combining improved SAE and Generalized Wiener process. On the one hand this paper utilizes separable CNN as the base neuron to construct a stacked self-encoder, which is combined with principal component analysis (PCA) to process the massive data and determine the HIs for uncertainty quantification of RUL prediction. On the other hand unlike the problem of ignoring the nonlinearity of the random error term in the existing degradation models, a generalized Wiener degradation process model with nonlinear drift is proposed to characterize the uncertainty of the RUL prediction. In order to realize adaptive prediction, the parameters of the Wiener degradation process model are estimated online by using the great likelihood estimation method combined with the Bayesian method. Finally, the effectiveness and superiority of the proposed method is verified through numerical simulation and practical case study of applied bearings.

The principal contributions of this study include: a separable CNN is used as the base neuron to construct a SAE,

which is combined with PCA to extract the deep hidden features reflecting the health state, while maintaining the topology and combinatorial relationship among the samples. An improved Wiener degradation process model with nonlinear drift is developed to make full use of the information in the HIs to update all the parameters in the Wiener degradation process model online. The presented approach can derive the probability density function while acquiring the RUL point estimate.

The remainder is organized as follows. Section 2 introduces a deep feature extraction technique that combines SAE and PCA. Section 3 uses a generalized Wiener degradation process model to construct HIs with adaptive updating of the RUL distribution. Section 4 utilizes numerical simulations and accelerated bearing life test data to verify the effectiveness and superiority of the proposed method. Section 5 concludes the study.

2. HI construction method based on SAE combined with PCA

2.1. Deep features extraction method based on SAE

SAE are deep neural networks composed of stacked layers of autoencoders [21]. The structure includes an input layer, layers of hidden neurons, and an output layer. The training process involves encoding and decoding phases. Input $X = (x_1, x_2, \dots, x_n)$, the encoding process transforms the input into an implicit state by means of the encoding function f ,

$$h = f(W \cdot X + b) \quad (1)$$

where W, b are the weight matrix and bias between the input and implicit layers, respectively. The decoding process reconstructs the implicit state h into $\bar{X}$ through the decoding function g ,

$$\bar{X} = g(W^* \cdot h + b^*) \quad (2)$$

where W^*, b^* are the weight matrix and bias between the implicit and output layers.

The network is trained by minimal reconstruction error, and when the reconstruction error is small enough, the implicit state h can be considered to contain all the information of the input X . The network is trained by reducing the reconstruction error. SAE adopts the idea of deep learning, which is more suitable for feature extraction by introducing multiple hidden layers and thus learning multiple intrinsic connections of the original data layer by layer. Figure 1 illustrates the architecture of the SAE network with two hidden layers. The SAE network exhibits a symmetric structure, where the output of the upper layer serves as the input for the lower layer. Once the raw data is input into the input layer, feature extraction begins in the first hidden layer. The resulting hidden state is then fed into the second hidden layer for further, deeper feature extraction. This process can be visualized as two encoders in series. Upon completion of the network training, it enables both data compression and deeper-level feature extraction.

Autoencoders can be constructed using various types of neural networks, such as multi-layer perception, LSTM networks, CNNs, and others. CNNs are widely used in the

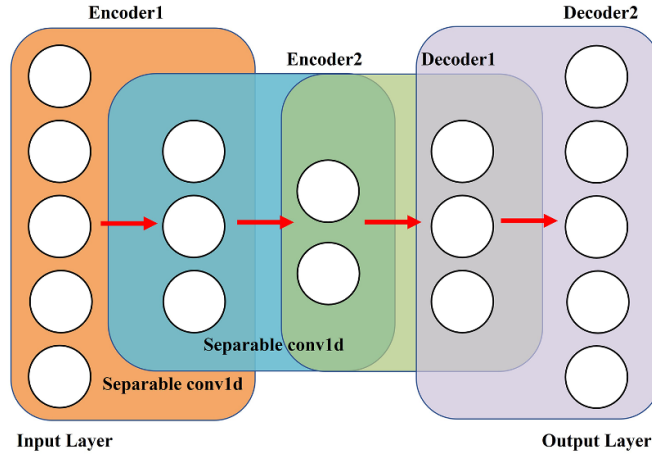


Figure 1. SAE network structure with two implicit layers.

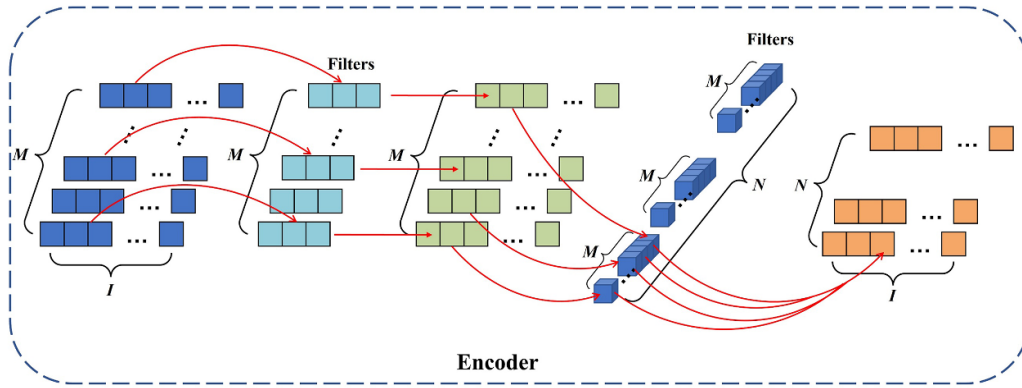


Figure 2. Single encoder structure schematic diagram.

field of mechanical equipment fault diagnosis. In contrast to other deep learning networks, they possess features like weight sharing, convolutional operations, and spatial pooling. These characteristics enable them to efficiently process high-dimensional raw data such as force signals, vibration signals, and acoustic emission signals. CNNs automatically extract degradation feature information from monitoring data [22]. Furthermore, Separable CNNs represent an improvement over traditional CNNs, comprising Depthwise Convolution and Pointwise Convolution layers. This structure, akin to traditional CNNs, is employed for feature extraction. However, in comparison to the conventional architectures, the number of parameters during model training significantly decreases. This reduction leads to lower computational demands and enhances the speed of model training [23, 24]. In Depthwise Convolution, each kernel is dedicated to a single channel, implying that each channel undergoes convolution with only one kernel. The number of kernels matches the number of channels in the previous layer, establishing a one-to-one correspondence between channels and kernels. The process of Pointwise Convolution closely resembles that of conventional convolution. The convolutional kernel size is $1 \cdot 1 \cdot M$, where M represents the number of channels in the

previous layer [25, 26]. Figure 2 illustrates the separable convolution to build a single Encoder structure.

It is worth noting that the improved SAE cannot maintain the topological and combinatorial relationships between samples, although it can extract deep features from the measurement data that respond to the health state. Therefore, in order to determine the final desired health metrics, it is necessary to introduce other downscaling methods to determine the health metrics, such as PCA, multidimensional scaling, Locally Linear Embedding, etc. The combination of SAE and PCA can be used to combine their strengths and determine the excellent health metrics.

2.2. HI determination

After extracting feature data from the raw data using SAE, the trend factor from the [27] is utilized as a quantitative representation to capture the trend of the extracted features. The trend factor serves as a crucial indicator to measure the relationship between feature extraction and time. Highly trending features were screened as inputs to PCA to support the identification of HIs. Suppose the extracted features are represented as $X = \{X(t_1), X(t_2), \dots, X(t_m)\}$, and the corresponding time

vector is denoted as $T = \{t_1, t_2, \dots, t_m\}$. The calculation of the trend factor is as follows,

$$V_{\text{corr}}(X, T) = \frac{\left| \sum_{k=1}^m (X(t_k) - \bar{X})(t_k - \bar{t}) \right|}{\sqrt{\sum_{k=1}^m (X(t_k) - \bar{X})^2 \sum_{k=1}^m (t_k - \bar{t})^2}} \quad (3)$$

where m is the number of samples, $\bar{X}$ and $\bar{t}$ are the mean values of X and $\bar{t}$, respectively.

PCA is a frequently employed linear dimensionality reduction technique [28, 29]. The fundamental concept involves transforming high-dimensional data into a lower-dimensional space through a linear projection. The objective is to capture the maximum information of the data in the projected dimensions, allowing the use of fewer dimensions while preserving more characteristics of the original data [30, 31]. Calculate the mean value of the N -dimensional feature matrix $F = [X_1, X_2, \dots, X_N]$

$$\mu = \frac{1}{N} \sum_{i=1}^N X_i. \quad (4)$$

Calculate the covariance matrix

$$\psi = \frac{1}{N} \sum_{i=1}^N (X_i - \mu)(X_i - \mu)^T. \quad (5)$$

Compute the eigenvalues and eigenvectors of the covariance matrix

$$\psi = EDE^T \quad (6)$$

where D is the diagonal array in descending order of eigenvalue size, $D = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_N)$, E is the set of eigenvalues λ_i corresponding to eigenvectors e_i and E is a standard orthogonal matrix, $E = (e_1, e_2, \dots, e_N)$.

PCA can transform the multidimensional features obtained from SAE into a one-dimensional HI, providing a direct representation of the true health condition of the equipment. To quantitatively characterize the prediction uncertainty of RUL, a method based on stochastic processes is introduced to construct a one-dimensional stochastic degradation model for the HI. This approach quantifies the prediction uncertainty.

3. RUL prediction based on generalized wiener process

3.1. Degradation process modeling

The Wiener process, as a non-monotonic degenerate process, has been widely used in the field of RUL prediction by virtue of its good mathematical properties. The degradation of the actual equipment at time t is represented by $X(t)$. It is

characterized using the following equation, incorporating a Wiener process with a nonlinear drift term:

$$X(t) = X(0) + \alpha \int_0^t u(\tau, \beta) d\tau + \sigma B(t) \quad (7)$$

where $X(0)$ denotes the initial degradation at time 0, $\alpha \int_0^t u(\tau, \beta) d\tau$ is the mean accumulated impact of the device degradation process, referred to as the nonlinear drift term $X(t)$, σ stands for the diffusion coefficient, and $B(t)$ represents standard Brownian motion. Specifically, α represents random parameters, while β and σ are common parameters. Without loss of generality, it is assumed that $X(0) = 0$ and $\alpha \sim N(\mu_\alpha, \sigma_\alpha^2)$, $B(t) \sim N(0, t)$. Statistical independence between α and $B(t)$ is assumed. The random error terms of the degradation model are all assumed to be $\sigma B(t) \sim N(0, \sigma^2 t)$. The random error terms can only characterize white noise or linear time-varying situations. However, due to the influence of environmental factors, degradation processes often involve non-stationary and nonlinear stochastic time-varying processes. Therefore, it is necessary to further extend the linear random error terms to more general nonlinear scenarios. Specifically for:

$$\begin{cases} X(t) = X(0) + \alpha \int_0^t u(\tau, \beta) d\tau + \varepsilon(t) \\ \beta \sim N(\mu_\beta, \sigma_\beta^2), \varepsilon(t) \sim N(0, (d_1 + d_2 t)^\gamma) \\ \text{Cov}\{\varepsilon(t_i), \varepsilon(t_j)\} = (d_1 + d_2 t)^\gamma, 0 \leq t_i \leq t_j \end{cases} \quad (8)$$

where $\varepsilon(t)$ is the random error term characterized by a nonlinear independent incremental process assuming the following properties:

- (1) $\{\varepsilon(t), t \geq 0\}$ is a continuous random process. $\varepsilon(t) \sim N(0, (d_1 + d_2 t)^\gamma)$ and $\varepsilon(t) \in (-\infty, +\infty)$.
- (2) For $0 \leq t_1 \leq t_2 \leq \dots \leq t_n$, $\varepsilon(t_1)$ is independent of the increments $\varepsilon(t_2) - \varepsilon(t_1)$, $\varepsilon(t_3) - \varepsilon(t_2)$, $\dots$, $\varepsilon(t_n) - \varepsilon(t_{n-1})$.
- (3) $\varepsilon(t_i)$ obeys the normal distribution $N(0, (d_1 + d_2 t)^\gamma)$. Increment $\varepsilon(t_i) - \varepsilon(t_{i-1})$ from the normal distribution $N(0, (d_1 + d_2 t_i)^\gamma - (d_1 + d_2 t_{i-1})^\gamma)$, $i = 1, 2, \dots, n$. moreover, the random variables β and $\varepsilon(t)$ are mutually independent.

In equation (8), $\alpha \sim N(\mu_\alpha, \sigma_\alpha^2)$ is used to describe the variability between individual devices. $u(\tau, \beta)$ represents the continuous drift function linked to a constant parameter β . When $u(\tau, \beta) = 1$, the model in this paper becomes a Wiener degenerate model with linear drift. When $d_1 = 0$, $\gamma = 1$, The model in this paper becomes the model in equation (7), which not only describes the case of nonlinear random errors, but also degenerates to the linear case.

3.2. RUL prediction

RUL can be interpreted as the duration until the HI $X(t)$ reaches the failure threshold D_f for the first time. The failure

threshold is often closely related to the specific technical indicators of the studied object or actual engineering system. To characterize the probability density function of the degraded device RUL, on the basis of the concept of the first hit time, the lifetime T of the degrading component is calculated

$$T = \inf \{t : X(t) \geq D_f | X(0) < D_f\} \quad (9)$$

where D_f represents the known failure threshold. Following the method outlined in [32] and considering the stochasticity of parameters, the probability density distribution function of the lifetime T defined by equation (9), can be expressed as:

$$f_T(t) = \frac{D_f - \alpha \int_0^t u(\tau, \beta) d\tau + \alpha(t, \beta)t}{\sqrt{2\pi(d_1 + d_2t)^\gamma t^2}} \times \exp \left\{ -\frac{\left[D_f - \alpha \int_0^t u(\tau, \beta) d\tau \right]^2}{2(d_1 + d_2t)^\gamma} \right\}. \quad (10)$$

Let $t_k (k \in N^+)$ represent the current time, and the RUL of the device at time t_k , denoted as L_k , can be expressed as:

$$L_k = \inf \{l_k : X(t_k + l_k) \geq D_f | X(t_k) < D_f\}. \quad (11)$$

Therefore, the probability density function of the RUL of the degrading device at time t_k can be expressed as:

$$f(l_k) = \frac{\gamma d_2}{(d_1 + d_2 l_k)^\gamma \sqrt{2\pi \left[(d_1 + d_2 l_k)^\gamma + \sigma_{\alpha,k}^2 \left(\int_{t_k}^{t_k+l_k} u(\tau, \beta) d\tau \right)^2 \right]}} \times \left\{ D_f - x_k - \left[\int_{t_k}^{t_k+l_k} u(\tau, \beta) d\tau - u(l_k + t_k, \beta) l_k \right] \right. \\ \times \frac{(D_f - x_k) \sigma_{\alpha,k}^2 \int_{t_k}^{t_k+l_k} u(\tau, \beta) d\tau + \mu_{\alpha,k} (d_1 + d_2 l_k)^\gamma}{\sigma_{\alpha,k}^2 \left(\int_{t_k}^{t_k+l_k} u(\tau, \beta) d\tau \right)^2 + (d_1 + d_2 l_k)^\gamma} \left. \right\} \\ \times \exp \left\{ -\frac{\left(D_f - x_k - \mu_{\alpha,k} \int_{t_k}^{t_k+l_k} u(\tau, \beta) d\tau \right)^2}{2 \left[\sigma_{\alpha,k}^2 \left(\int_{t_k}^{t_k+l_k} u(\tau, \beta) d\tau \right)^2 + (d_1 + d_2 l_k)^\gamma \right]} \right\} \quad (12)$$

where $\mu_{\alpha,k}$ and $\sigma_{\alpha,k}^2$ are the mean and variance, respectively, of the random parameter α obtained by updating at moment t_k . x_k is the degraded observation of the degraded device at moment t_k . When the relevant parameters ($\mu_{\alpha,k}$, $\sigma_{\alpha,k}^2$, β , d_1 , d_2 , γ) in the model are known, by introducing equation (12), the RUL can be predicted adaptively.

The equation provided offers an approximate analytical representation of the PDF of the RUL, accounting for the influence of random parameters. The point estimate of the RUL can be derived from the analytical expression of the PDF as its corresponding expectation:

$$E(l_k) = \int_{-\infty}^{+\infty} l_k f(l_k) dl_k. \quad (13)$$

3.3. Estimation of initial values of parameters by maximum likelihood estimation

Assuming that there are M similar devices with degraded data, note that the number of degraded data for the i th device is N_i . Let $X(t_{i,j})$ (abbreviated as $X_{i,j}$) denote the j th degraded data of the i th device at time $t_{i,j}$, and $1 \leq j \leq N_i$. Based on equation (8), it can be obtained that

$$X_{i,j} = X(0) + \alpha_0 \int_0^{t_{i,j}} u(\tau, \beta) d\tau + \varepsilon(t_{i,j}) \quad (14)$$

where α_0 denotes the prior value of the random parameter α and its distribution is $\xi_0(\alpha) \sim N(\mu_{\alpha,0}, \sigma_{\alpha,0}^2)$. If the degraded data of the i th degraded device is $X_i = (X_{i,1}, X_{i,2}, \dots, X_{i,N_i})^T$, then X_i obeys a multivariate Gaussian distribution with mean and covariance as follow, respectively

$$\mu_i = \mu_{\alpha,0} I_i \quad (15)$$

$$\Sigma_i = \sigma_{\alpha,0}^2 I_i I_i^T + K_i \quad (16)$$

where $I_i = (\int_0^{t_{i,1}} u(\tau, \beta) d\tau, \int_0^{t_{i,2}} u(\tau, \beta) d\tau \dots \int_0^{t_{i,N_i}} u(\tau, \beta) d\tau)^T$,

$$K_i = \begin{bmatrix} (d_1 + d_2 t_{i,1})^\gamma & (d_1 + d_2 t_{i,1})^\gamma & \dots & (d_1 + d_2 t_{i,1})^\gamma \\ (d_1 + d_2 t_{i,1})^\gamma & (d_1 + d_2 t_{i,2})^\gamma & \dots & (d_1 + d_2 t_{i,2})^\gamma \\ \vdots & \vdots & \ddots & \vdots \\ (d_1 + d_2 t_{i,1})^\gamma & (d_1 + d_2 t_{i,2})^\gamma & \dots & (d_1 + d_2 t_{i,N_i})^\gamma \end{bmatrix}.$$

The log-likelihood function produced from the deteriorated data of the M impaired devices is

$$\ell(\Theta | X_{1:M}) = -\frac{1}{2} \ln(2\pi) \sum_{i=1}^M N_i - \frac{1}{2} \sum_{i=1}^M \ln |\Sigma_i| \\ - \frac{1}{2} \sum_{i=1}^M (X_i - \mu_i)^T \Sigma_i^{-1} (X_i - \mu_i). \quad (17)$$

The optimal value of the unknown parameter Θ is obtained by the maximization of the log-likelihood function equation (18). Next, real-time updating of the distribution parameters of the stochastic parameter α can be achieved at any moment t_k using the online observation data x_k .

3.4. Stochastic parameters are updated in real time

For a particular degraded device, at any moment t_k during its lifetime, the stochastic parameter α of the degradation model can be estimated from all the observations $x_{1:k} = \{x_1, x_2, \dots, x_k\}$ of the device up to the moment t_k . With the help of a Bayes rule, the posterior distribution of the stochastic parameter α of this degradation model can be obtained at the moment t_k [33]:

$$p(\alpha | x_{1:k}) \propto p(x_{1:k} | \alpha) \xi_0(\alpha) \quad (18)$$

$$p(x_{1:k}|\alpha) = \frac{1}{\prod_{q=1}^k \sqrt{2\pi [(d_1 + d_2 t_q)^\gamma - (d_1 + d_2 t_{q-1})^\gamma]}} \times \exp \left\{ - \sum_{q=1}^k \frac{\left[x_q - x_{q-1} - \alpha \int_{t_{q-1}}^{t_q} u(\tau, \beta) d\tau \right]^2}{2 [(d_1 + d_2 t_q)^\gamma - (d_1 + d_2 t_{q-1})^\gamma]} \right\} \quad (19)$$

where $p(x_{1:k}|\alpha)$ denotes the likelihood function given the random parameter α . The prior distribution $\xi_0(\alpha)$ has been estimated by equation (17). Based on equations (18) and (19), the mean and variance of $p(\alpha|x_{1:k})$ are obtained, respectively,

$$\mu_{\alpha,k} = \frac{\frac{\mu_{\alpha,0}}{\sigma_{\alpha,0}^2} + \sum_{q=1}^k \left\{ \frac{(x_q - x_{q-1}) \int_{t_{q-1}}^{t_q} u(\tau, \beta) d\tau}{(d_1 + d_2 t_q)^\gamma - (d_1 + d_2 t_{q-1})^\gamma} \right\}}{\sum_{q=1}^k \left\{ \frac{\int_{t_{q-1}}^{t_q} u(\tau, \beta) d\tau}{(d_1 + d_2 t_q)^\gamma - (d_1 + d_2 t_{q-1})^\gamma} \right\}} + \frac{1}{\sigma_{\alpha,0}^2} \quad (20)$$

$$\sigma_{\alpha,k}^2 = \frac{1}{\sum_{q=1}^k \left\{ \frac{\int_{t_{q-1}}^{t_q} u(\tau, \beta) d\tau}{(d_1 + d_2 t_q)^\gamma - (d_1 + d_2 t_{q-1})^\gamma} \right\}} + \frac{1}{\sigma_{\alpha,0}^2} \quad (21)$$

The time indicator corresponds one-to-one with the acquired monitoring data, which means that the length of the monitoring data is equal to the length of the time indicator. The time index is represented as $t_{1:k} = \{t_1, t_2, t_3, \dots, t_k\}$.

The estimated parameters derived from equations (17), (20) and (21) are utilized in equation (12) to derive the probability density function for the remaining lifespan of the device at a given time. The key procedural steps are outlined as follows:

- Step 1: Offline parameter estimation. This is done by utilizing historical degradation data from M similar devices to estimate the hyperparameters in the prior distributions of both common and stochastic parameters using equation (17).
- Step 2: Online parameter update. For a specific in-service equipment, the parameters $\mu_{\alpha,k}$ and $\sigma_{\alpha,k}^2$ are updated in real time using equations (20) and (21) after observing the degradation data x_k at any time t_k during its life cycle.
- Step 3: Online prediction of RUL. the parameter estimates obtained in Steps 1 and 2 are brought into equation (12) to obtain the probability density function of the device at any moment t_k the probability density function of RUL.

Step 4: Obtain the observation data x_{k+1} of the degraded device at time t_{k+1} , repeat steps 2 and 3 to obtain the probability density function of the device RUL at time t_{k+1} .

The flowchart for HI estimation and RUL forecast is presented in figure 3. The first section uses offline data to learn the network parameters, and once the learning is completed, the real-time data features are fed into the combination of SAE and PCA to obtain new HI. The second part of the online update includes the Wiener process parameters and the probability distribution of the RUL.

4. Case studies

4.1. Numerical simulation

To verify that the model proposed in this paper can solve the problem that the existing model ignores the random term error as nonlinear, the proposed model is considered to be compared with the model of equation (7). For convenience, M0 and M1 are used to denote the methods in this paper and the comparison methods, respectively. Due to the advantages of power functions in describing mechanical equipment degradation, in the simulations and case studies, the drift coefficient $\int_0^t u(\tau, \beta) d\tau = t^\beta$ in the HI model based on the Wiener degradation process is used. The following stochastic degradation model is employed to generate simulated data:

$$X(t) = X(0) + \alpha \int_0^t u(\tau, \beta) d\tau + \varepsilon(t). \quad (22)$$

Set the values of the simulation parameters as $\mu_\alpha = 1$, $\sigma_\alpha^2 = 0.062$, $\beta = 1.5$, $d_1 = 0.2$, $d_2 = 0.3$, $\gamma = 0.05$. According to the above model description and settings, a total of 10 degradation trajectories are generated, as shown in figure 4, and one degradation trajectory is arbitrarily selected for the simulation study.

The trajectory contains a total of 22 degradation data, the maximum value of which is $X(22) = 290$. The degradation failure threshold is set to $D_f = 290$, and the corresponding moment of the trajectory that first reaches the failure threshold is 22. The remaining portion of the degradation trajectory serves as historical data for estimating the initial values of the unidentified parameters in the degradation model. The results of the fitting process are presented in table 1.

Two methods are used to fit the degradation trajectory data, and figure 5 shows the prediction of degradation in line with the Wiener process, which shows that the suggested method can better reflect the degradation process.

To illustrate the validity and reasonableness of the methodology of this paper, the estimated values of the model parameters are brought into equation (12) to obtain the predicted value and PDF of RUL, and to further evaluate the suggested methodology in comparison with the M1 methodology, the average

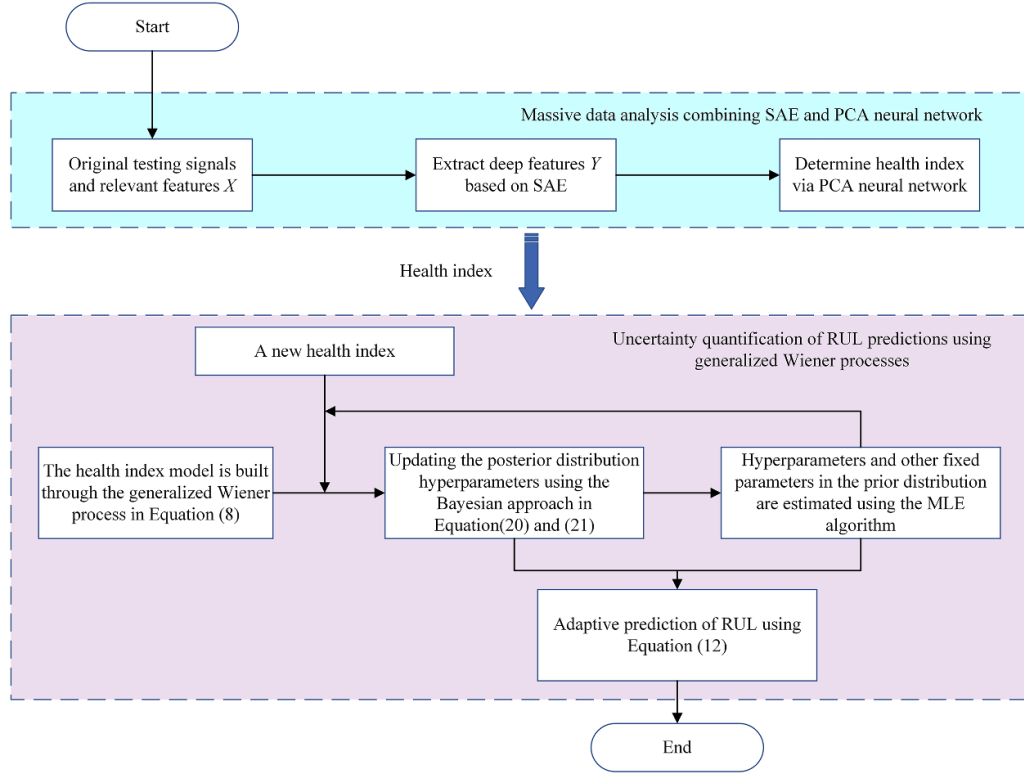


Figure 3. Flowchart of the proposed method.

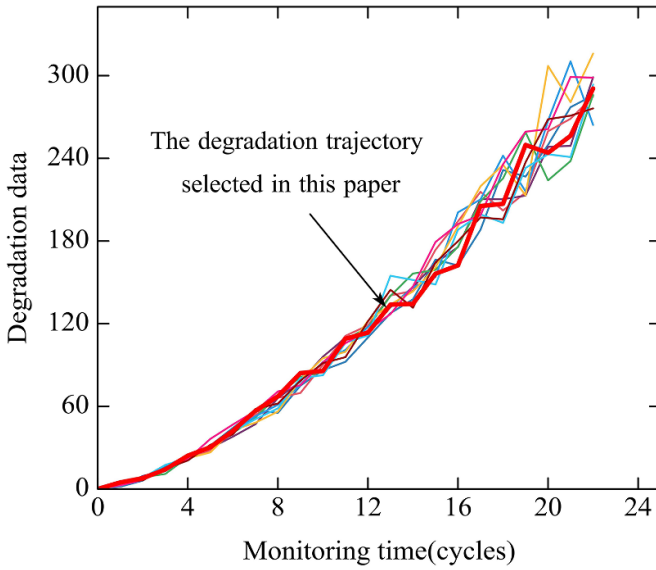


Figure 4. Simulation of degradation trajectories.

relative error indicator is introduced, which is defined as follows:

$$\text{MRE} = \frac{1}{N} \sum_{k=1}^N \frac{|\bar{l}_k - l_k|}{\bar{l}_k} \quad (23)$$

where $\bar{l}_k$ and l_k are the true and predicted RUL of the equipment at time t_k , respectively, and N is the total number of predicted

Table 1. *A priori* parameter estimates for M0 and M1.

Method	$\mu_{\alpha,0}$	$\sigma_{\alpha,0}$	β	d_1	d_2	γ
M ₀	1.357	0.00 550	1.594	0.192	0.246	0.0474
M ₁	2.0121	0.00 210	1.624	0	0.0421	1

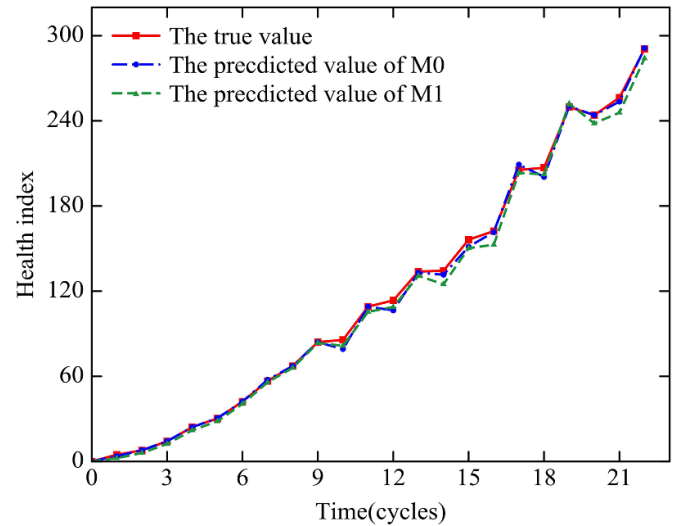


Figure 5. Degradation trajectory prediction.

moments. This metric reflects the average value of the relative error of prediction over the life cycle. The prediction results are shown in figure 6.

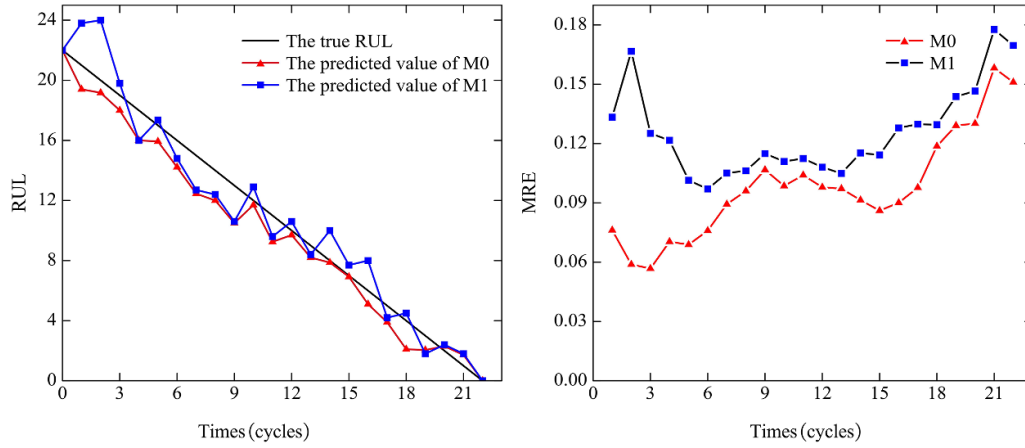


Figure 6. Comparison of RUL prediction results.

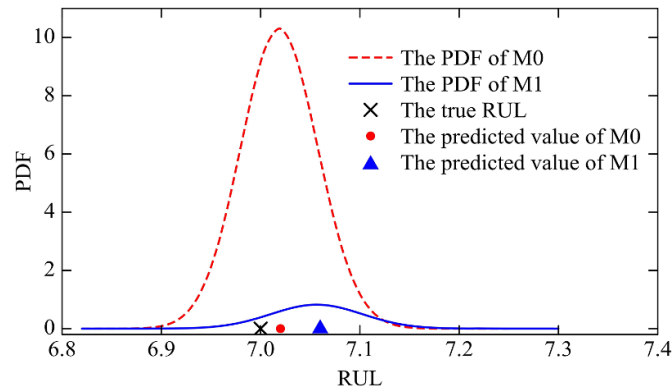


Figure 7. PDF curves of RUL for M0 and M1 methods at moment 15.

Overall, the prediction accuracy of the M0 method is better than that of the M1 method, which proves that the method proposed in this paper can better solve the effects brought by non-linear random errors, estimate the parameters more accurately, and then improve the prediction accuracy. The RUL probability density functions predicted by the two methods at the 15th moment are displayed in figure 7. From the figure, it can be observed that the PDFs of RUL corresponding to M0 and M1 can contain the real RUL of the equipment, but the PDF curve corresponding to the method proposed in this paper is obviously narrower, higher and more concentrated, and is closer to the real life of the equipment, which indicates that the prediction accuracy of the proposed model is better than that of M1.

4.2. Experimental demonstration

4.2.1. Introduction of the experimental platform. The experimental data from the 2012 PHM Challenge was utilized to validate and demonstrate the effectiveness of the prediction model presented in this study. The experimental data on bearing were acquired from the FEMTO-ST. As shown in figure 8, the test platform mainly consists of a rotating part, a loading part and a measuring part.

The measurement setup consists of two accelerometers and a temperature sensor. The accelerometers are mounted

radially on the bearing outer ring. The temperature sensor is located in a hole near the outer bearing collar. Since the vibration signals obtained from accelerometers can serve as indicators of the bearing's operational condition, the analysis primarily focuses on the vibration signals from the accelerometers. The acceleration sensor records data every 10 s for 0.1 s, with a sampling frequency of 25.6 kHz. Bearing degradation experiments were conducted on 17 sets of bearings under three different operating conditions. Condition 1 (1800 rpm and 4000 N) included bearings 1_1–1_7, Condition 2 (1652 rpm and 4200 N) included bearings 2_1–2_7, and Condition 3 (1500 rpm and 5000 N) included bearings 3_1–3_3. In order to avoid propagation of damages to the whole test bed (and for security reasons), tests were stopped when the amplitude of the vibration signal overpassed 20 g. The training and test sets were set by the organizers. Bearings 1_1, 1_2, 2_1, 2_2, and 3_1 in the dataset are used as the training set, utilizing full lifespan vibration data. Bearings 1_3, 1_4, 1_5, 2_3, 2_4, 2_5, and 3_1 are used as the test set, utilizing truncated lifespan data.

4.2.2. HI determination. To fully utilize the feature information present in the vibration signals, the time–frequency domain characteristics of the vibration signals are derived

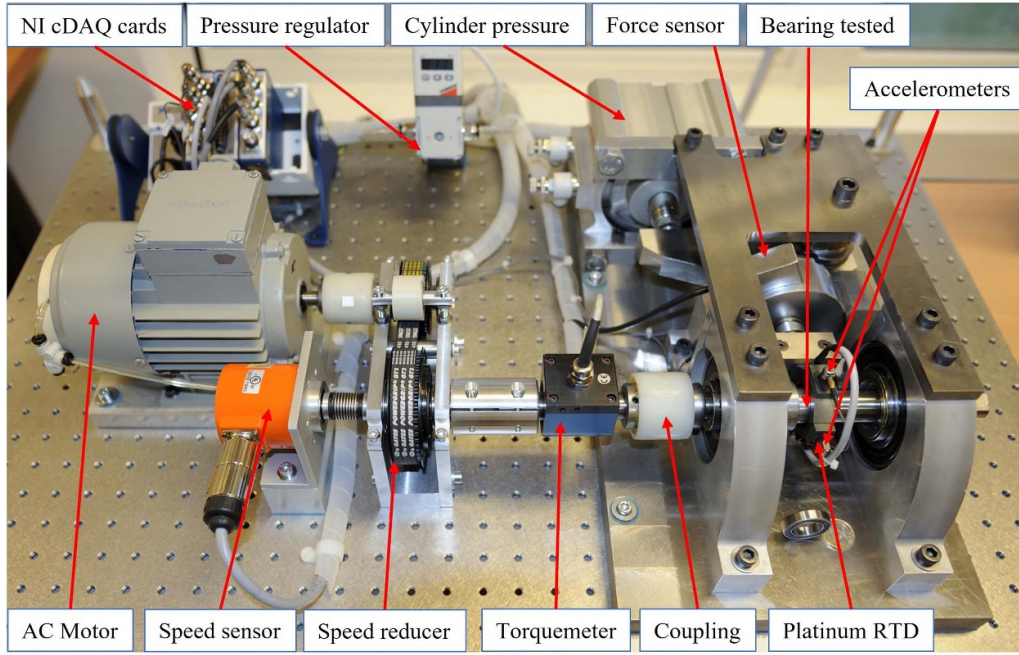


Figure 8. PRONOSTIA experimental platform.

Table 2. Stacked self-encoder network architecture for bearing data sets.

Layer	Input dimension	Output dimension	Dilation rate	Activation	Convolution kernel
Encoder layer	2560	640	1	Sigmoid	3×1
Encoder layer	640	10	2	Sigmoid	3×1
Decoder layer	10	640	4	Sigmoid	3×1
Decoder layer	640	2560	8	Sigmoid	3×1

through the application of the short-time Fourier transform. These characteristics are then utilized as input for the enhanced stack self-encoder to facilitate unsupervised feature extraction. The two-layer stacked self-encoder structure is 2560-640-10-640-2560, and the encoder and decoder each comprise two layers, with dimensions based on central symmetry. Which reduces the original signal features to 10 dimensions, in which the separable CNN kernel size is set to 3, and the activation function is sigmoid. The network structure is shown in table 2. The features with higher trend are selected and input into PCA to obtain the one-dimensional HIs. Taking the bearing in Condition 1 as an example, the bearing HI obtained by the suggested method is displayed in figure 9.

To demonstrate the superiority of the combined SAE and PCA method, SAE, PCA, and MDS techniques were introduced to reduce the 2560-dimensional features to 1-dimensional features for comparison. The trend factor in

equation (3) is used as a measure of the excellence of the HI, and the results of the comparison of the trend factors of the four methods for the HI of each bearing are given in table 3. The results show that, except for bearing 2_1, the trend factors of the HI extracted by the suggested method for the rest of the bearings are larger than those of the other methods.

4.2.3. RUL prediction. Based on the receipt of the bearing HI, in order to visually describe the RUL prediction uncertainty, bearing1_3 and bearing2_7 are selected as representatives to achieve the HI modeling and RUL prediction, and the HIs are shown in figure 10.

To assess the RUL prediction capabilities of the proposed method, six cycles from bearings 1_3 and 2_7 were chosen as the prediction periods for quantitative analysis. The expected value and PDF of RUL can be obtained using equations (16) and (17), respectively. In addition, the M1 method was implemented to establish the HI model and the RUL prediction for comparative purposes.

Figures 11 and 12 present the PDF curves and corresponding predicted expectations of RUL for the selected prediction periods using Method M0 and Method M1. From the graphs, it is evident that the PDF curves of both methods encompass the actual RUL. However, as the data accumulates, the RUL predicted by Method M0 becomes closer to the real values, and the prediction uncertainty gradually decreases.

Figure 13 shows the prediction results for bearing 1_3 and bearing 2_7 to visually compare the accuracy of the two prediction methods. The proposed method is more accurate.

Further quantification of the prediction results of the two models, taking bearing 2_7 as an example, this study employs

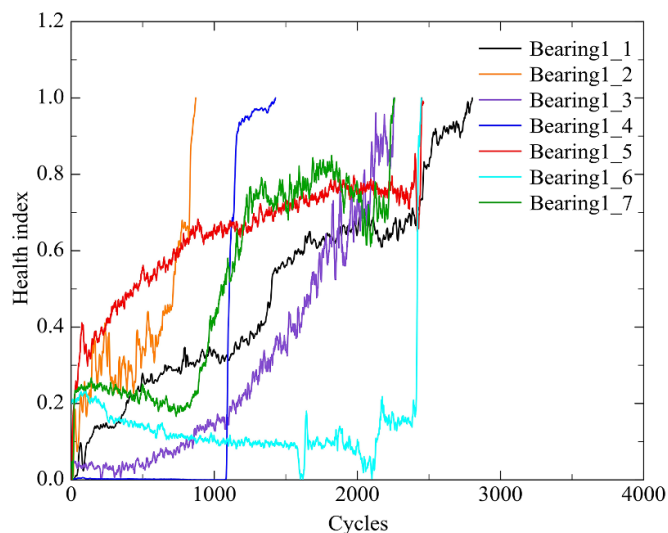


Figure 9. Bearing health indicator.

Table 3. The trend factor comparison of the HI.

Testing bearing	Proposed method	SAE	PCA	MDS
Bearing1_1	0.9788	0.7756	0.4479	0.4612
Bearing1_2	0.8407	0.3523	0.3421	0.3409
Bearing1_3	0.8716	0.2426	0.3177	0.3661
Bearing1_4	0.7374	0.7339	0.7105	0.6743
Bearing1_5	0.8670	0.3331	0.0202	0.1393
Bearing1_6	0.8873	0.3456	0.1270	0.1645
Bearing1_7	0.8769	0.2018	0.3300	0.2281
Bearing2_1	0.5848	0.6149	0.2470	0.2105
Bearing2_2	0.8462	0.8389	0.6557	0.2949
Bearing2_3	0.7384	0.4471	0.1674	0.1146
Bearing2_4	0.8947	0.2919	0.2959	0.1739
Bearing2_5	0.7698	0.7569	0.6309	0.1455
Bearing2_6	0.9099	0.5886	0.1461	0.2176
Bearing2_7	0.9259	0.2643	0.2885	0.2522
Bearing3_1	0.5691	0.3386	0.3797	0.1514
Bearing3_2	0.6443	0.4016	0.3219	0.1738
Bearing3_3	0.8282	0.6636	0.8036	0.6969
Mean	0.8101	0.4818	0.3665	0.2827

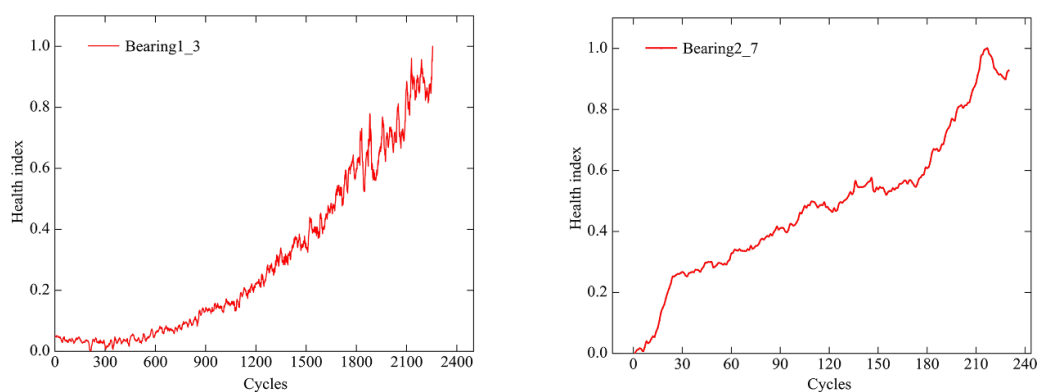


Figure 10. The health indicator.

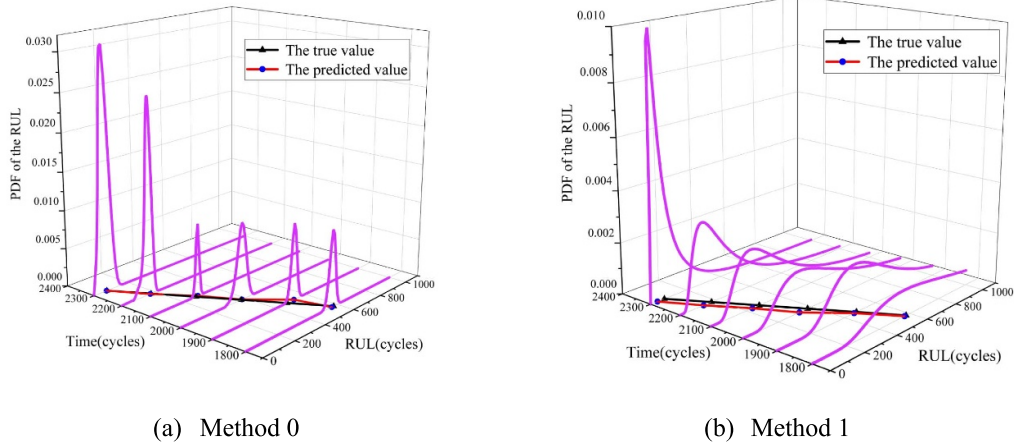


Figure 11. PDFs of RULs for bearings 1_3.

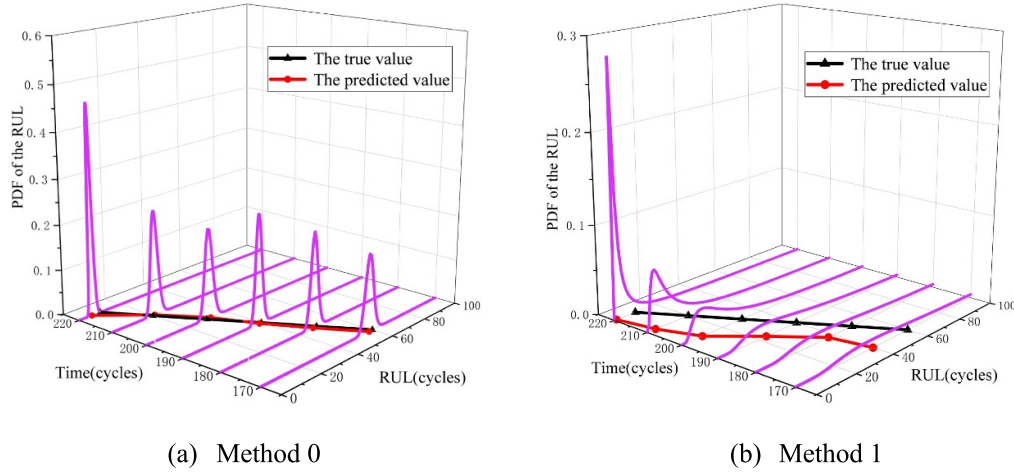


Figure 12. PDFs of RULs for bearings 2_7.

common performance metrics in the reliability domain: absolute error and relative error indices. For the i th test dataset, the relative error between the actual RUL and the predicted RUL value can be represented as:

$$RE_i = \frac{RUL_i - ActRUL_i}{ActRUL_i} \cdot 100\%. \quad (24)$$

Based on the data presented in figure 14 and table 4, it is evident that the M0 method demonstrates an enhanced ability to accurately estimate parameters and subsequently improve the accuracy of predictions as degradation data accumulates.

To illustrate the effectiveness of the suggested approach, a comparative analysis was conducted using methods from the following references to predict the RUL of the studied bearings. Specifically, the model-based method from [34] is referred to as M2, [35] is referred to as M3, [36] is referred to as M4, and [37] is referred to as M5. Additionally, the

performance of the prediction methods was comprehensively evaluated using the scoring function from the IEEE PHM 2012 Challenge [38]. The corresponding expression for evaluation is as follows:

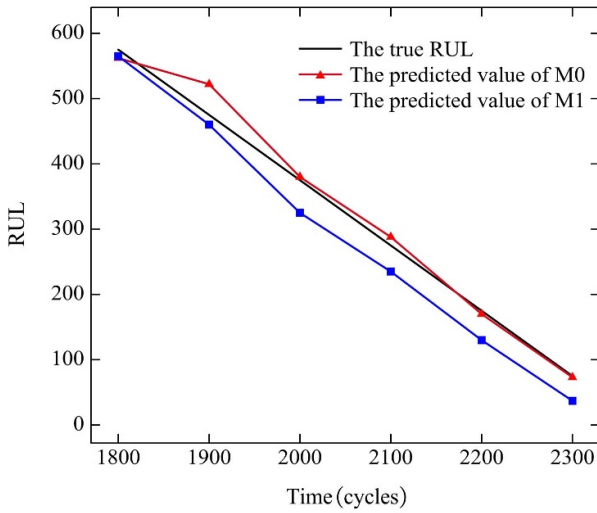
$$Score = \frac{1}{5} \sum_{i=1}^5 A_i \quad (25)$$

$$A_i = \begin{cases} \exp(-\ln(0.5) \cdot (Er_i/5)), & \text{if } Er_i \leq 0 \\ \exp(+\ln(0.5) \cdot (Er_i/20)), & \text{if } Er_i > 0 \end{cases} \quad (26)$$

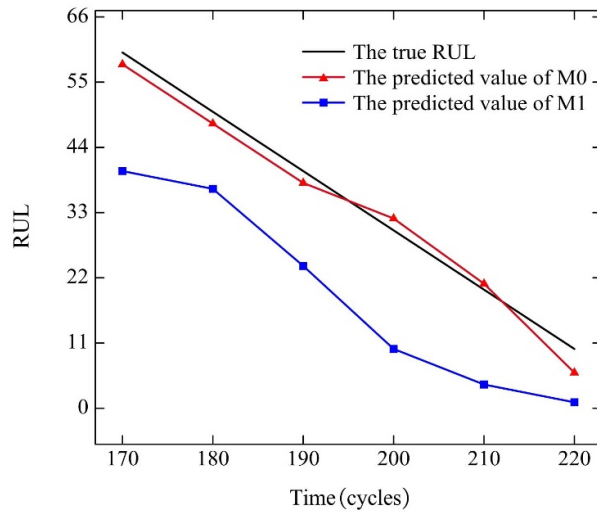
where Er_i denotes the error percentage for the i th test data set

$$Er_i = \frac{ActRUL_i - RUL_i}{ActRUL_i} \cdot 100 \quad (27)$$

where $ActRUL_i$ and RUL_i represent the actual RUL and the predicted RUL of the i th test dataset, respectively. In addition, other performance metrics are selected for further comparison,



(a) Comparison of prediction results for bearing 1_3



(b) Comparison of prediction results for bearing 2_7

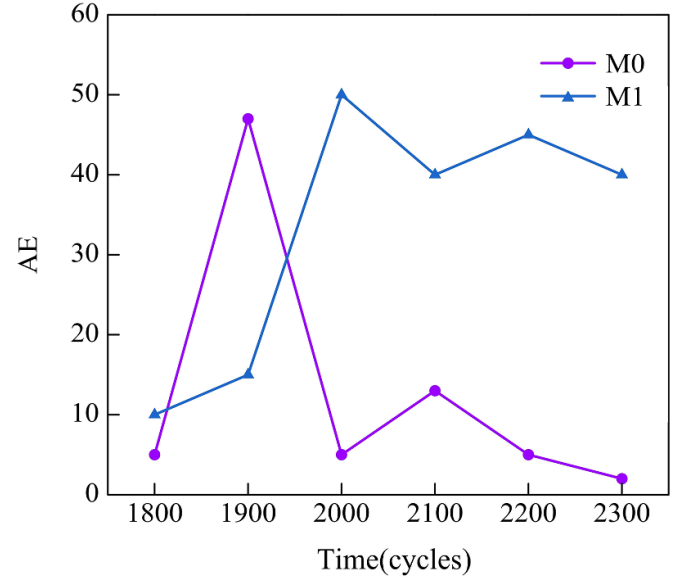
Figure 13. Comparison of the prediction results of the two methods for different bearings.

the mean absolute error (MAE) and the normalized root mean square error (NRMSE), denoted as:

$$MAE = \frac{1}{5} \sum_{i=1}^5 |ActRUL_i - RUL_i| \quad (28)$$

$$NRMSE = \frac{\sqrt{\frac{1}{5} \sum_{i=1}^5 (ActRUL_i - RUL_i)^2}}{\left(\frac{1}{5} \sum_{i=1}^5 RUL_i \right)}. \quad (29)$$

The prognostication outcomes for RUL and the relative error of bearings by the suggested method and other four

**Figure 14.** Bearing 2_7 absolute error of RUL prediction for both methods.**Table 4.** Results of comparison of relative errors.

Method	Relative errors at different monitoring points					
	170	180	190	200	210	220
M0	-3.3%	-4.0%	-2.5%	6.0%	5.0%	-40.0%
M1	-33.3%	-26.0%	-40.0%	-33.3%	-80.0%	-90.0%

methods are presented in table 5. Table 5 compares the performance metrics of different methods. As can be seen from table 5, although the relative errors of the suggested method for bearings 1_5 and 1_7 are not the smallest in the corresponding comparisons, the proposed method is closer to the actual RUL of the bearings on the whole, and these errors are completely acceptable in practice. Moreover, in most of the test sets, the predicted RUL of the planned method is smaller than the actual RUL, which is conducive to the subsequent development of maintenance decisions.

Furthermore, the performance metrics in table 6 reflect the scores, MAE, and NRMSE of the proposed method. It is worth noting that since the literature on Method 4 did not provide the RUL for bearings 1-3, 1-4, and 1-5, their performance metrics are not calculated. The proposed method's NRMSE is not the lowest value, because that NRMSE is not only related to the deviation between the actual RUL and the predicted value of the RUL, but also to the predicted value of the RUL. Consequently, the NRMSE is only capable of partially reflecting the discrepancy between the predicted and actual RUL values. In instances where the predicted value of the RUL is relatively low, the NRMSE may be larger. Although the proposed method's performance is not the best, other methods can only obtain point estimates of RUL without assessing its

Table 5. RUL prediction results and relative error.

Test bearings	Current time (cycles)	Actual RUL (cycles)	Method 0	Method 2	Method 3	Method 4	Method 5
1_3	1802	573	570 (−0.52%)	558 (−2.62%)	582 (1.57%)	—	592 (3.32%)
1_4	1139	289	243 (−15.92%)	190 (−34.25%)	350 (21.11%)	—	339 (17.30%)
1_5	2302	161	293 (81.98%)	328 (103.7%)	120 (−25.47%)	—	392 (143.47%)
1_6	2302	146	125 (−14.38%)	92 (−36.99%)	204 (39.72%)	178 (21.92%)	183 (25.34%)
1_7	1502	757	644 (−14.93%)	706 (−6.74%)	659 (−12.95%)	778 (2.78%)	867 (14.53%)

Table 6. The performance metric comparisons of different methods.

Performance metric	M0	M2	M3	M5
Score	0.56	0.30	0.38	0.17
MAE	63.00	70.00	53.00	89.40
NRMSE	0.22	0.25	0.16	0.25

uncertainty. The proposed method helps quantify prediction uncertainty.

5. Conclusion

An adaptive prognostic technique is suggested for forecasting the RUL by integrating the SAE and Generalized Wiener Process to take advantage of the enormous amount of data available during the life cycle. Firstly, the deep hidden features of the bearings are extracted using SAE. Following this, the input of PCA is filtered to include only the features with high trend, which are then used to generate HIs reflecting the bearings' health status. Secondly, a HI evolution model for generalized Wiener degradation processes is proposed and PDFs for RUL prediction is conceptually derived based on the first hitting time. Finally, the results of numerical simulation studies show that the proposed model RUL prediction accuracy is improved while reducing the uncertainty. The results of the experimental study of the bearing degradation data show that the trendiness of the extracted HIs reaches 0.81, which is higher than that of several other methods, and the performance indicators predicted by RUL have a Score of 0.56, a MAE of 63.00, and a NRMSE of 0.22, which demonstrates the ability to accurately predict RUL. It shows the validity and superiority of the proposed method.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: www.femto-st.fr/.

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Conflict of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRedit authorship contribution statement

Zhe Chen: Investigation, Methodology, Software, Validation, Writing—original draft. **Yonghua Li:** Investigation, Writing—review & editing, Supervision. **Qi Gong:** Validation, Writing—review & editing. **Denglong Wang:** Data curation, Methodology. **Xuejiao Yin:** Writing—review & editing.

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